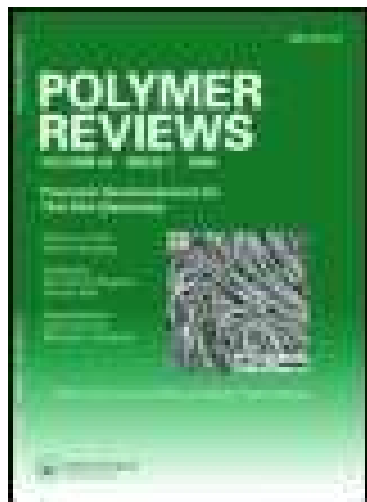




Polymer Reviews

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/lmsc20>

A Review of the Properties and Applications of Poly (Methyl Methacrylate) (PMMA)

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Published online: 24 Jun 2015.



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To cite this article: Umar Ali, Khairil Juhanni Bt. Abd Karim & Nor Aziah Buang (2015): A Review of the Properties and Applications of Poly (Methyl Methacrylate) (PMMA), Polymer Reviews, DOI: [10.1080/15583724.2015.1031377](https://doi.org/10.1080/15583724.2015.1031377)

To link to this article: <http://dx.doi.org/10.1080/15583724.2015.1031377>

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A Review of the Properties and Applications of Poly (Methyl Methacrylate) (PMMA)

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Advances in the use of poly (methyl methacrylate) (PMMA) have opened up a wide range of applications in the field of nanotechnology. The knowledge of the properties of PMMA has contributed a lot to the recent boosts in the synthesis, modification, and applications of the polymer. However, there is a need to condense these developments in the form of an article for better understanding and easy access. This review highlights the fundamental physical properties of PMMA, coupled with experimental evidence of its essential chemistry, such as solubility, hydrolysis, grafting, combustion reactions, reactions with amines, and thermal decomposition. The recent developments in the applications of PMMA in biomedical, optical, solar, sensors, battery electrolytes, nanotechnology, viscosity, pneumatic actuation, molecular separations, and polymer conductivity were also revealed.

Keywords poly (methyl methacrylate), physical properties, applications, nanotechnology, chemical properties, polymer

1. Introduction

Poly (methyl methacrylate) (PMMA), having the IUPAC name of poly [1-(methoxy carbonyl)-1-methyl ethylene] from the hydrocarbon standpoint, and poly (methyl 2-methyl-propenoate) from the ester standpoint, is a synthetic polymer from the methyl methacrylate monomer as illustrated in Figure 1.¹ PMMA was discovered in the early 1930s by British chemists, Rowland Hill and John Crawford, followed by its first application by a German chemist, Otto Rohm, in 1934.² PMMA is an optically clear (transparent) thermoplastic, and it is widely used as a substitute for inorganic glass, because it shows high impact strength, is lightweight, shatter-resistant, and exhibits favorable processing conditions.³ Outstanding properties include weather resistance and scratch resistance. The presence of the adjacent methyl group (CH₃) in the polymer structure prevents it from packing closely in a crystalline fashion, and from rotating freely around the C-C bonds. This is why PMMA was found to be an amorphous thermoplastic. The first major application of the polymer took place during World War II, when PMMA was used as aircraft windows and bubble canopies for gun turrets.

Received December 18, 2014; accepted March 2, 2015.

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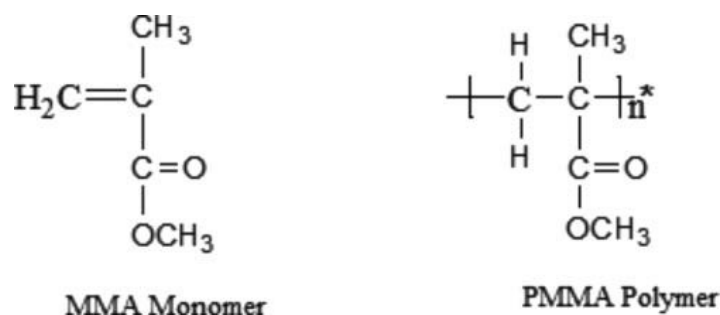


Figure 1. The structures of MMA and PMMA.

PMMA is a promising polymer for applications in optical, pneumatic actuation, sensor, analytical separation, and conductive devices.^{4–8} Other applications include the use of PMMA in biomedical applications, polymer electrolytes, polymer viscosity, and drug delivery using electro-diffusion or electro-osmotic flow.^{9–13} Due to its compatibility and easy processing as a polymer moiety, PMMA with carbon nanotubes or other inorganic materials plays an important role in the development of nanotechnology. Wang et al., in the preparation of carbon nanotube polymer composites, used poly (styrene-co-acrylonitrile) with poly (methyl methacrylate)-g-multi walled carbon nanotubes.¹⁴

PMMA is the polymer that enhanced the behavior at the interface between the carbon nanotube and the copolymer.

PMMA can be obtained from its monomer using different techniques of polymerization. The monomer undergoes polymerization using the common methods of free radical and anionic initiations by bulk, solution, suspension, and emulsion techniques.¹⁵ Following the discovery of a new technique of polymerization by Krzysztof Matyjaszewski in 1995,¹⁶ called Atom Transfer Radical Polymerization (ATRP), Matyjaszewski et al. successfully polymerized the monomer of methyl methacrylate (MMA) to produce PMMA as a living polymer with 80% conversion, polydispersity as low as 1.1, and an Mn of 20,000 in a few hours.¹⁷

In this review, we adequately condensed the physical and chemical properties of PMMA as a synthetic polymer. Emphasis was given to identifying the properties that influenced the wider applications of the polymer. Finally, this paper reveals the recent advancements in the applications of the polymer in the fields of biomedical, molecular separation, solar, sensor, battery electrolytes, pneumatic actuation, polymer conductivity, viscosity, and nanotechnology applications. In view of this, this review may open a gateway to researchers in polymer science and technology, and assist industrialists in actualizing the experimental facts into large scale manufacturing.

2. Properties

From the chemistry point of view, elements, compounds, molecules, and macromolecules possess some qualities that prove their existence as entities to scientists or analysts. Examples of such qualities are color, odor, texture, boiling point, melting point, density, crystallinity, elasticity, and many more, depending on the substance in question. This is what scientists refer to as the physical properties of a substance or material. However, the qualities possessed by these substances can be altered, modified, or permanently changed to another form, with the intention of producing a new substance with new qualities. This can only be achieved through the interactions

of the substance with another substance of the same or different type. This is what scientists refer to as the chemical properties of a substance. Therefore, PMMA as a macromolecule cannot be an exception; it has both physical and chemical properties that shall be discussed.

2.1. Physical Properties of PMMA

PMMA is one of the amorphous polymers that belongs to the acrylate family. It is a clear, colorless polymer with a glass transition temperature range of 100°C to 130°C, and a density of 1.20 g/cm³ at room temperature. This polymer melts at 130°C, with a water absorptivity of 0.3%, moisture absorption at an equilibrium of 0.3 to 0.33%, and a linear shrinkage mold of 0.003 to 0.0065 cm/cm¹⁸⁻²⁰.

PMMA is among the polymers that have high resistance to sunshine exposure, because it has a small variation under the effect of UV-radiation. It has very good thermal stability, and is known to withstand temperatures as high as 100°C and as low as -70°C. It also possesses very good optical properties, with a refractive index of 1.490, and a good degree of compatibility with human tissue.^{18,19}

In the field of mechanical strength, PMMA has a high Young's Modulus and a low elongation at breakage. Therefore, it does not shatter upon rupture, and happens to be one of the hardest thermoplastics with high scratch resistance. This polymer has a reasonable resistance to chemicals, being unaffected by the aqueous solution of most laboratory chemicals. However, it has a low resistance to chlorinated and aromatic hydrocarbons, esters, or ketones.^{18,19}

Polymer tacticity is the relative stereochemistry of adjacent chiral centers within a polymer, especially vinyl polymers. The physical properties of a polymer, such as glass transition temperature (T_g), solubility, crystallinity, and hydrolysis, depend not only on the type of monomer comprising it, but also on the secondary and tertiary structures, that is, the stereochemistry of the linkage.²¹⁻²⁶ Based on this fundamental chemistry, PMMA could be isotactic, syndiotactic, or atactic.

Isotactic (mm) is when the adjacent monomer groups were added in the meso diad mode, and the ester groups on the successive asymmetric carbons are projected on the same side of the polymer chain. Syndiotactic (rr) is when the monomer groups were added in the racemic diad mode, and the ester groups on the successive asymmetric carbons are projected in a regular alternation on both sides of the plane in the polymer chain. Atactic (mr) is another racemic diad mode, but the ester groups on the successive asymmetric carbons are projected randomly on both sides of the plane in the polymer chain as shown in Figure 2.^{22,25}

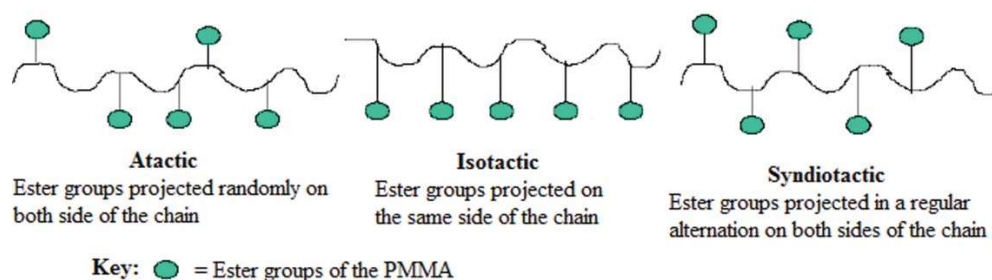


Figure 2. An illustration of the different tacticities of PMMA.

Pure PMMA could be synthesized in isotactic, atactic, and syndiotactic forms using control/living radical polymerization, reversible addition-fragmentation chain transfer (RAFT), and anionic polymerization, based on the judicious choice of initiator, solvent, and monomer feed.^{22,23} The amorphous nature of PMMA is governed by its tacticity in the following order: syndiotactic > atactic > isotactic. The T_g goes in descending order: $130^\circ\text{C} > 120^\circ\text{C} > 55^\circ\text{C}$, respectively.²¹

Pure PMMA could be in the forms of isotactic, syndiotactic, and atactic tacticity when produced by radical and anionic polymerization, with the judicious choice of initiator, solvent, and monomer feed.^{22,23} Commercial PMMA is the cross-linking mixture of isotactic PMMA/syndiotactic PMMA, atactic PMMA/sPMMA, and aPMMA/iPMMA. Sometimes, a copolymer of plasticizer/PMMA, with the highest percentage of PMMA, is used just to make it more flexible for wider applications. Therefore, the properties of commercial PMMA, such as glass transition temperature (T_g), solubility, and level of crystallinity, vary widely.^{21, 25}

Due to its physical properties illustrated in Table 1, PMMA has very wide applications in the areas of science and technology, such as synthesis, composites, medical, optical, nanotechnology, and analytical separations, just to name a few. Therefore, the physical properties of any substance are a simple guide revealing the potentials of the substances in terms of the chemistry and applications.

2.2 Chemical Properties of PMMA

2.2.1 Solubility. In general, solubility is the ability of a solute (solid, liquid, or gas) to dissolve in a solvent (liquid or gas) to form a homogenous solution (mixture) of the solute and solvent. Basically, the solubility of a substance depends on both the physical and chemical properties of the solute and the solvent, as well as the temperature.

Considering simple inorganic molecules, such as salts, their solubility is faster due to quick solvolysis by the solvent on the ions of the solute because they are mostly ionic in nature. On the other hand, for organic compounds, their solubility is governed by a popular aphorism of “like-dissolve-like.” This means that a solute will dissolve best in a solvent that has a similar chemical structure to itself. In this situation, the solvation capacity of the solvent primarily depends on its polarity. PMMA is an organic polymer, and its solubility is expected to be governed by “like-dissolve-like,” with polarity playing a major role.

PMMA shows little deviation as its solubility is more complex, starting with swelling in the solvent and the subsequent formation of a very soft layer on its surface. This is then followed by diffusion of the solvent into the whole polymer before it gives a homogenous solution with the solvent. This is the reason why PMMA takes a few minutes before it is dissolved completely, even if it is in its best solvent.

Considering the applications of PMMA, which lead to its exposure to chemicals in different forms, there is a need for a comprehensive guide on the chemical resistance of PMMA.^{31–33} Chemicals can affect the strength, flexibility, and surface appearance of polymeric materials due to: (i) attack on the polymer chain and (ii) absorption, resulting in swelling and dissolution. These attacks depend on factors such as temperature, pressure, time, and stress.³⁴ Tables 2–7 show a guide to the chemical durability/resistance of PMMA.

Table 1
Physical and some mechanical properties of PMMA.

Property	PMMA	References
Color	Courless	19, 27
Density (g/cm ³)	1.18	25, 28
Melting Point (°C)	220–240	27
Surface Hardness (Rockwell)	M92, M90-M100	27, 28
Glass Transition Temp. (T _g)	110 to 120	19, 25
Linear Thermal Expansion ($\times 10^{-5}$ mm/mm.k)	6.3	28
Thermal Expansion Coefficient 20°C ($\times 10^{-4}$ K ⁻¹)	1.8	25
Thermal Expansion Coefficient 40°C ($\times 10^{-4}$ K ⁻¹)	2.2	25
Isothermal Compressibilities at 20°C ($\times 10^{-5}$ bar ⁻¹)	2.7	25
Isothermal Compressibilities at 40°C ($\times 10^{-5}$ bar ⁻¹)	2.9	25
Mold Shrinkage (%)	0.3–0.6	27
Water Absorption (%)	0.3	27
Oxygen Index (%)	19, 17.19	27, 29
Thermal conductivity at 20°C (k(W/mK))	0.12, 0.17	25, 30
Tensile Strength (MPa)	72	28
Tensile Modulus (GPa)	3.10	28
Elongation at Break (%)	5	28

Table 2
PMMA: Resistance to organic solvents.

PMMA			
Chemical	20°C, 30 Days	50°C, 30Days	References
Acetone	SA	SA	34, 35
Benzene	SA	SA	34, 35
Chloroform	SA	SA	34, 35
Cyclohexane	LA	-	34, 35
Diethylene Glycol	SA	SA	34, 35
Dimethyl Formamide (DMF)	SA	SA	34, 35
Dimethyl Sulfoxide (DMSO)	-	-	34, 35
Dioxane	SA	SA	34, 35
Ethyl acetate	SA	SA	34, 35
Ethylene Glycol	NA	LA	34, 35
Glycerol	NA	-	34, 35
Naphthalene	NA	LA	34, 35
Tetrahydrofuran (THF)	SA	SA	34, 35
Trichloroethane	SA	SA	34, 35
Toluene	SA	SA	34, 35
Turpentine	LA	SA	34, 35

Key: NA = Not Attacked LA = Lightly Attacked SA = Strongly Attacked.

Table 3
PMMA: Resistance to acids and bases.

PMMA			
Chemical	20°C, 30 Days	50°C, 30 Days	References
Acetic Acid (10%)	NA	LA	34, 35 31
Acetic Acid (100%)	SA	SA	34, 35 31
Formic Acid (10%)	NA	LA	34, 35
Hydrochloric Acid	NA	NA	34, 35
Hydrofluoric Acid	SA	SA	34, 35
Lactic Acid	NA	NA	34, 35
Nitric Acid (10%)	NA	-	34, 35
Nitric Acid (Conc.)	SA	SA	34, 35
Sulfuric Acid (10%)	NA	NA	34, 35 32
Sulfuric Acid (90%)	SA	SA	34, 35 32
Tartaric Acid	NA	LA	34, 35
Sodium hydroxide	NA	NA	34, 35 31
Sodium hydroxide (Conc.)	LA	SA	34, 35 31
Potassium hydroxide	NA	-	34, 35
Potassium hydroxide (Conc.)	LA	SA	34, 35

Key: NA = Not Attacked; LA = Lightly Attacked; SA = Strongly Attacked

The above-mentioned data have been obtained by test; therefore, the same results may not be obtained using different conditions.³⁵

The solubility of polymers in ionic liquids (ILs) is a recent development in polymer solubility. This may be due to the nonpolarity of the ionic liquids, which is the predominant solvent parameter of the ILs. Ionic liquids could be cationic or anionic structure dependent, and include but are not limited to bis(trifluoromethanesulfonyl)amide-1-alkyl-3-methylimidazole ($C_4mimNTf_2$), N-butyl-N-methylpyrrolidone bis(trifluoromethanesulfonyl)amide ($P14NTf_2$) for cationic structure dependence, and hexafluorophosphate-1-

Table 4
PMMA: Resistance to alcohols.

PMMA			
Chemical	20°C, for 30 Days	50°C, for 30 Days	References
Benzyl Alcohol (pure)	SA	SA	34, 35
Ethanol (30%)	LA	SA	34, 35
Methanol	-	-	34, 35
Methyl Alcohol (10%)	NA	LA	34, 35
Methyl Alcohol (pure)	SA	SA	34, 35
Phenol	SA	SA	34, 35
Propyl Alcohol	NA	LA	34, 35

Key: NA = Not Attacked; LA = Lightly Attacked; SA = Strongly Attacked.

Table 5
PMMA: Resistance to oils and fats.

PMMA			
Chemicals	20°C, for 30 Days	50°C, for 30 Days	References
Butyl stearate	NA	NA	34, 35
Coconut Oil	NA	NA	34, 35
Lanolin	NA	NA	34, 35
Mineral Oils	NA	LA	34, 35
Paraffin Oil	NA	NA	34, 35

Key: NA = Not Attacked; LA = Lightly Attacked; SA = Strongly Attacked.

Table 6
PMMA: Resistance to food products.

PMMA			
Chemicals	20°C, for 30 Days	50°C, for 30 Days	References
Fruit Juice	NA	NA	34, 35
Milk	NA	NA	34, 35
Olive Oil	NA	NA	34, 35
Vinegar	NA	NA	34, 35
Wine	NA	NA	34, 35

Key: NA = Not Attacked; LA = Lightly Attacked; SA = Strongly Attacked.

alkyl-3-methylimidazole (C_4mimPF_6 , trifluoromethanesulfonate-1-alkyl-3-methylimidazole ($C_4mimTfO$) for anionic structure dependence.³⁶

Ueno et al. reported the solubility of PMMA in ionic liquids using PMMA-grafted silica nanoparticles in the presence of different cationic and anionic structures. They empirically revealed that PMMA was soluble in hydrophobic ILs, such as ($C_4mimNTf_2$) and (C_4mimPF_6).³⁶

The dissolution of polymers in ionic liquids is a complex phenomenon that begins with the swelling of the polymer in the IL, by absorbing and holding the IL in the polymer network, resulting in a polymer/IL binary system (ion gels). This opens a gateway for new applications of the ion gels as polymer electrolytes in electrochemical devices.

Evchuk et al. studied the solubility of PMMA in 12 organic solvents in order to reveal the possible correlation between the physico-chemical characteristics of organic solvents and the rate of polymer dissolution in them. They used a temperature range of 30–70°C and the following solvents: benzene, toluene, o-xylene, m-xylene, trichloromethane, trichloroethylene, 1, 4-dioxane, cyclohexanone, acetophenone, ethyl acetate, pentyl acetate, and dimethyl formamide. From the results, it was found that there is no direct correlation between the dissolution rate and the structure or properties of the solvents, including polarity. Trichloroethylene was found to be the best solvent for PMMA, while trichloromethane was the poorest. Polar solvents, such as ethyl acetate and

The deviation from the normal exhibited by PMMA in this study may be attributed to the high molecular weight of the polymer and chain crosslinking. In addition, the researchers did not take into consideration the tacticity of PMMA, the isotactic and syndiotactic arrangement in Figure 3, which has a great impact on the properties of the polymer.

It helps to understand the thermal behavior dissolution in its solvent and its mechanical properties. However, it was found practically that tetrahydrofuran (THF) was capable of dissolving PMMA at room temperature quicker than most of the 12 organic solvents used by Evchuk and his co-workers. Further investigation can be done on the solubility of PMMA, with respect to its tacticity, for more clarification.

2.2.2 Acid Hydrolysis. PMMA hydrolyzed completely with sulfuric acid (H_2SO_4) to become poly (methacrylic acid) (PMAA). The isotactic and syndiotactic PMMA behaved differently in sulfuric acid hydrolysis. Isotactic PMMA hydrolyzed to PMAA completely at 55°C for 1 h, while the syndiotactic PMMA at the same temperature hydrolyzed (limitedly) to 85% in more than 24 hrs. In addition, the syndiotactic PMMA hydrolysis did not follow the equilibrium trend of low molecular weight esters, because it was found to be unaffected by changes in the concentration of the water and by-product alcohol.^{23,26} Hydrochloric acid and hydro-iodic acid are capable of hydrolyzing PMMA, but at a slow rate when compared to sulfuric acid.

2.2.3 Combustion Reaction. PMMA has a predominantly elemental composition of carbon and hydrogen. Therefore, it is liable to undergo an exothermic combustion reaction to yield gaseous products (CO_2 , CO , H_2O ,) and energy like any other hydrocarbon. However,

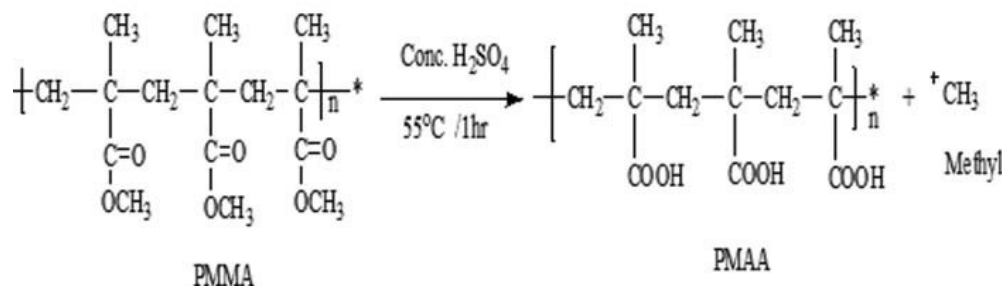
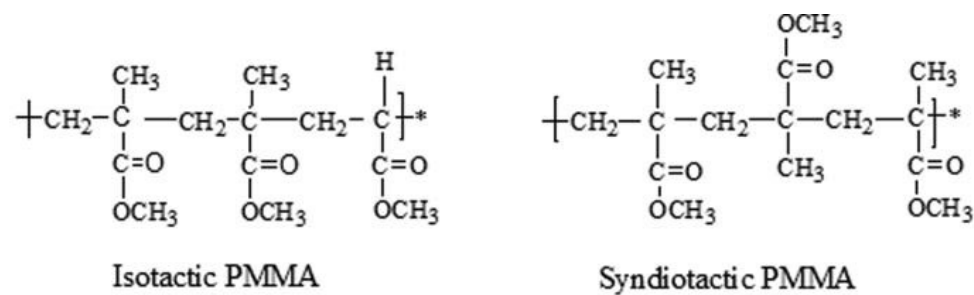


Figure 3. The acid hydrolysis of isotactic PMMA.

the combustion reaction of PMMA is a complex process that involves three (3) major stages due to its properties as a thermoplastic macromolecule:

1. The decomposition (depolymerization) of the PMMA to generate the monomer (MMA).
2. The monomer (MMA) further decomposes to yield small molecular gaseous products, such as methane, methanol, formaldehyde, etc.
3. Finally, the small gaseous products undergo combustion to yield the final products, such as CO_2 , CO , H_2O , and energy.³⁸

This explains why all thermoplastic polymers shrink and melt when in contact with direct flames before they ignite. Moreover, they take a longer combustion time when compared to non-thermoplastics like paper and cotton fabrics. This is due to the formation of an interface between the condensed phase and the gaseous phase.

2.2.4 Thermal Decomposition. The thermal decomposition of PMMA has been extensively studied in the absence of oxygen. The decomposition temperature varies, depending on the approach used in the synthesis of the polymer. Radically polymerized PMMA containing terminal C—C bonds decomposes at a temperature of 220°C with simple mechanisms of monomer repeat units bond scission and C-C bond random scission as in Figure 4. The anionically polymerized PMMA containing saturated end groups decomposes at a higher temperature, with random C-C bond scission initiation. However, in both cases, the predominant product is the monomer of MMA with a greater yield.^{19,38}

In the presence of oxygen, PMMA undergoes complex thermal oxidative decomposition, due to the reaction of the macro-radicals with oxygen. The products of the decomposition at 260°C involve the formation of 2-methyl-oxirane, carbonic acid, methyl ester, and dimethyl itaconate. However, the predominant product is also the monomer of MMA, with a 95.5% yield. Generally, the main product of the air decomposition of PMMA at a temperature of 300°C was the monomer of MMA. Other small molecules, such as CH_3OH , CH_4 , CO_2 , CO , and H_2O , were generated as a result of the β scission of the ester group of the PMMA, with further reactions as illustrated below in (1), (2), (3), and (4), respectively.²⁹

Considering the temperatures of both the thermal and thermally oxidative decompositions of PMMA, the polymer can hardly decompose at a temperature below 200°C . This

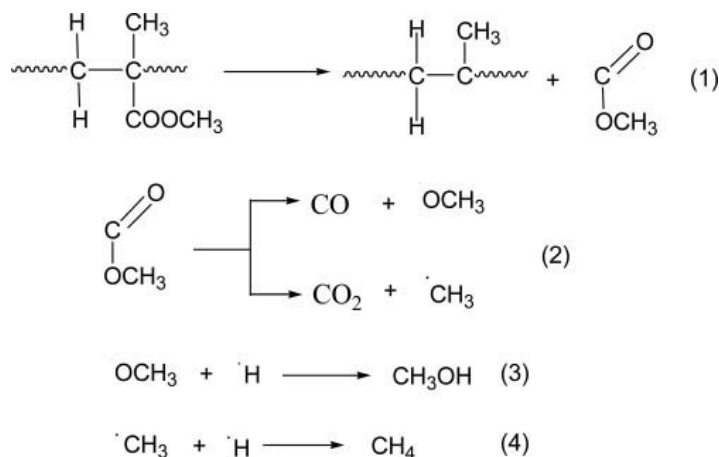


Figure 4. Thermal decomposition Mechanism of PMMA.

Table 7

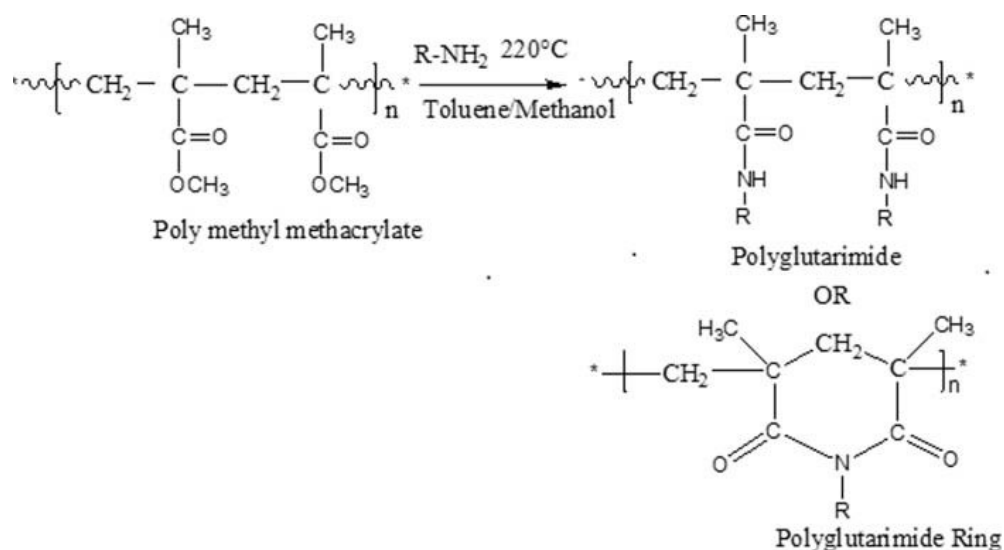
The content of thermal decomposition products of PMMA in air at different temperatures.

Products	Content v/v%	
	300°C	500°C
MMA	95.5	78.9
CH ₄	0.8	1.3
CH ₃ OH	1.8	3.2
CO ₂	0.8	6.0
CO	0.2	0.3
H ₂ O	0.4	4.5

is the basic reason why the polymer has a very good thermal stability and weather resistance as two of its outstanding properties.

The v/v% unit was used because the products are in gaseous form and gas chromatography was the instrument used for the analysis.²⁹

2.2.5 Reaction with Amines. PMMA as a macromolecule is capable of reacting with primary amines at the terminal methoxy group of the polymer. Chin et al. were the first to perform this type of reaction, as reported in their work for the synthesis and characterization of poly (glutarimide) (PGI). The synthesis was achieved by dissolving the PMMA in a solvent mixture of toluene and methanol (ratio 9:1), followed by a subsequent reaction with various primary amines in an autoclave at 200°C for 3 hrs. In the course of the imidization of the polymer, there is the possibility of having a poly (glutarimide) ring as one of the products, and crosslinking as a side reaction, which may reduce the percentage yield of PGI as shown in Figure 5. Therefore, the amount of PMMA with respect to the solvent was reduced to minimize the side reactions.³⁹ The conversion of PMMA to poly (glutarimide) opens a gateway for subsequent research in the modification of the polymer for other applications (Figure 4).

**Figure 5.** The conversion of PMMA to polyglutarimide.

Bulmuş et al. reported the modification of PMMA monosized microbeads for the immobilization of glucose oxidase. Polyvinyl alcohol was used in the synthesis of the polymer to introduce a hydroxyl group, which opens the door for a further conversion to aldehyde and reaction with hexamethylene diamine for aminated PMMA.⁴⁰

Fixe et al. reported the functionalization of PMMA using hexamethylene diamine to yield primary amino groups at the surface of the polymer for the suitable immobilization of biomolecules like DNA. This was done by the available methyl ester of the polymer, under basic pH conditions, with an electron donor (N) present on the primary amine.⁴¹

It has been proven beyond a reasonable doubt that PMMA's capability of interacting with amines can be regarded as one of its potentials in biomedical applications. Therefore, it appears to be an opportunity for researchers to conduct more scientific investigations using the PMMA polymer and other protein macromolecules besides DNA for more applications in bio-molecular analyses. This could be advantageous due to the low cost of the polymer.

2.2.6 Grafting/Crosslinking. Grafting is the process of generating macromolecules called copolymers, which involve two dissimilar monomers or polymers that are covalently bonded. In the situation that involves two or more dissimilar polymers, one of the polymers must act as a trunk polymer (backbone). Crosslinking is the multidirectional interaction of monomers or polymers for chain extension, leading to the formation of network structures. Crosslinking results due to the polymerization of polymers or monomers with functionalities greater than 2, and is accomplished by chemical reaction, an increase in temperature, or both. The mechanistic approaches of grafting and crosslinking are basically divided into three (3) based on the approach of the interactions: polymer/polymer (P/P) approach, polymer/monomer (P/M) approach, and monomer/monomer (M/M) approach.⁴² PMMA, due to its reactivity, compatibility, and ease of modification, undergoes grafting/crosslinking using all three (3) mechanistic approaches.

2.2.6.1 Polymer/Polymer Mechanistic Approach. This type of mechanistic approach is about forming a copolymer through covalent and other crosslinking bonds (hydrogen bond and other intermolecular bonds) via the interaction between two different types of polymers. Jin et al. prepared a composite of multi-walled carbon nanotubes by the melt blending method. The carbon nanotubes were found to be well dispersed in the PMMA matrix at the temperature of 200°C. The TGA, TEM, and DMA confirmed the successful preparation of the composites at the above temperature without causing damage to the nanotube or the polymer.⁴³

A similar interaction was reported by Krüchel et al. when they produced a PMMA composite filled with carbon black and carbon fiber. The two composites were subjected to oscillatory shear deformations, and their electrical conductivity was measured. The composites of carbon fiber were found to exhibit a high percolation threshold and low conductivity when compared to the composites of carbon black.⁴⁴ Sometimes, the polymer/polymer mechanism can be achieved via direct chemical reaction to form a covalent bond between the polymers involved, in the presence of a solvent.

Work reported by Baskaran et al. illustrated this scenario. They used a "grafting to" strategy involving hydroxyl-terminated PMMA and acyl chloride functionalized multi-walled carbon nanotubes in a toluene solvent. The grafted copolymer (PMMA-g-MWCNT) was characterized using spectroscopic, microscopic, and thermo-gravimetric analyses.⁴⁵

The two common strategic approaches for the preparation of copolymers and composites are melt blending and direct chemical reaction using solvents. Melt blending is more

advantageous than the direct chemical reaction because the copolymers or composites produced are free from contamination by residual solvents. However, it has the limitation of generating a weak bond at the polymer to polymer interface when compared to the direct chemical reaction strategy.

2.2.6.2 Polymer/Monomer Mechanistic Approach. A polymer and monomer interaction to produce a copolymer is a mechanism in which the polymer may act as a living polymer or macro-initiator. However, in some cases, the polymer and the monomer can be copolymerized using chemical initiators. The monomer undergoes *in situ* polymerization during the process initiated by the polymer (macro-initiator) or chemical initiator, generating the copolymer through covalent bond formation. This mechanism of grafting for PMMA was found to have the highest number of reports in the literature.

In 2013, Rani et al. reported the synthesis of PMMA grafted sodium alginate using ceric ammonium nitrate (CAN) as a free radical initiator. The grafting process was assisted by the microwave irradiation of the mixture of sodium alginate and PMMA for 3 minutes. The characterization of the copolymer was done by FT-IR, SEM, and TGA.⁴⁶

Using a similar initiator of ceric ammonium nitrate, Routray et al. grafted PMMA onto cellulose acetate in the presence of dimethylsulfoxide (DMSO) as a solvent. Different parameters, such as grafting yield, grafting efficiency, and total conversion of the monomer to a polymer, were evaluated at different reaction conditions of temperature, time, monomer concentrations, and initiator concentrations. The grafted product was characterized using FT-IR and ^1H -NMR.⁴⁷ Wang et al. enhanced the mechanical properties of poly (styrene-co-acrylonitrile) using the copolymer of PMMA grafted multi-walled carbon nanotubes, as reported in their work.

The synthesis of PMMA-g-MWCNT illustrates a typical case of using a macro-initiator in the grafting mechanism. A macro-initiator of the brominated multi-walled carbon nanotube (MWCNT-Br) initiated the atom transfer radical polymerization (ATRP) that led to the formation of the copolymer of PMMA-g-MWCNT.¹⁴ Generally, carbon nanotubes (CNTs) have very limited solubility because of the high tendency toward aggregation due to the π - π bond interaction. However, the attachment of a polymer moiety by grafting onto the CNTs increases its solubility, especially in organic solvents. This happens to be a breakthrough in the exploitation of CNT potentials in terms of application.

Harish et al. reported that the grafting of PMMA from chitosan produced the copolymer of chitosan-g-PMMA films, by copolymerization initiated by potassium persulfate ($\text{K}_2\text{S}_2\text{O}_8$). The work focused much attention on the biodegradation of the copolymer by the use of screened microorganisms capable of biodegradation. The characterization of the copolymer and biodegraded samples was carried out using FT-IR, SEM, and HPLC.⁴⁸ Chitosan is known to be soluble in protonating solvents, especially acids, but chemistry-wise there may be an improvement in the solubility extended to some non-protonating solvents as a result of the interaction between the methoxy groups of PMMA and the amine group of the chitosan, instead of the hydroxyl groups. However, there is a need for further investigations to confirm the idea.

2.2.6.3 Monomer/Monomer Mechanistic Approach. The prerequisite conditions of generating a copolymer using the monomer/monomer mechanism is that the two monomers must be different or multifunctional otherwise, leading to the formation of a homopolymer, not a copolymer. This mechanism is similar to the conventional polymerization mechanism, but differs in the final products obtained due to crosslinking, branching, and block copolymer formation.

Barroso et al. illustrated the scenario of the monomer/monomer mechanism in the synthesis of a copolymer of PMMA and PMAA from their monomers in supercritical

carbon dioxide (scCO₂) using a 2, 2'-azobis (isobutyronitrile) (AIBN) initiator. The copolymer was characterized by FT-IR, ¹H-NMR, GPC, and SEM. The same product was used in developing a pH-responsive membrane by the scCO₂-assisted phase inversion method.⁴⁹

High refractive-index trialkoxysilane-capped PMMA-titania optical thin films were prepared by Lee and his co-workers using THF as a solvent. The first step in the preparation of the optical thin films is the interaction between the MMA and 3-(trimethoxysilyl) propyl methacrylate (MSMA) monomers, in the presence of benzyl peroxide (BPO) as an initiator. The product of the reaction is the copolymer of the two monomers, which acts as an intermediate for the generation of the final product.⁶

Nien et al. introduced PMMA-acrylic acid sodium salt particles into bone cement, which is capable of the total release of certain amounts of drugs without losing the mechanical properties of the cement. Three monomers, MMA, acrylic acid (AA), and allylmethacrylate (AMA), were cross-linked by free radical polymerization using a 2, 2'-azobis (isobutyronitriles) (AIBN) initiator to produce the copolymer. The copolymer incorporated into the commercial bone cement was found to increase the hydrophilicity of the cement. This allows fluids to pass through the cement more efficiently, and to supplement both the drug release rate and the total release of the drugs.⁵⁰

Considering the above literature reports, they show the similarities of the mechanism with conventional polymerization, since both use thermal initiators, but lead to the formation of different products of copolymers, respectively.

3. Applications of PMMA

3.1 Biomedical Applications

PMMA has been used in the area of biomedical applications, which involves the preparation of bone cements for drug delivery/release and cranioplasty (Figure 6). The qualities that made the polymer a potential material for these applications include: non-toxicity, less cost, easy processability, compatibility, minimal inflammatory reactions with tissues, and greater fracture resistance, especially when used in cranioplasty.

PMMA has also been used to widen the applications of chitosan in various fields that include biomedical and pharmaceutical applications. Zuhair et al. reported the successful grafting of a PMMA/chitosan blend. The results indicated an increase in the mechanical properties, such as tensile strength and flexural modulus. The degradation, porosity, and water absorbency of the blend in synthetic body fluid (SBF) with a pH of 7.4 increased with an increase in the chitosan percentage and immersion time in SBF. These behaviors exhibited by the PMMA/chitosan blend illustrate its potentials for drug release applications.⁵¹

Polyurethane-PMMA (PU-PMMA) can be used for textile applications; however, the incorporation of TiO₂ into the structure is expected to give excellent biocompatibility and other related properties.

Zuber et al. communicated the preparation of PU-PMMA/titanium oxide (PU-PMMA/TiO₂) blend base composites. Based on the results of the investigations made, the blend was nontoxic to living cells, had good mechanical strength, and biocompatibility for applications in dental implants.⁵²

Among other reports in the literature is work reported by Nein et al. They synthesized bone cement for drug release with the capability of control, and an increase in both the drug release rate and total release amount of the drugs, without losing the mechanical

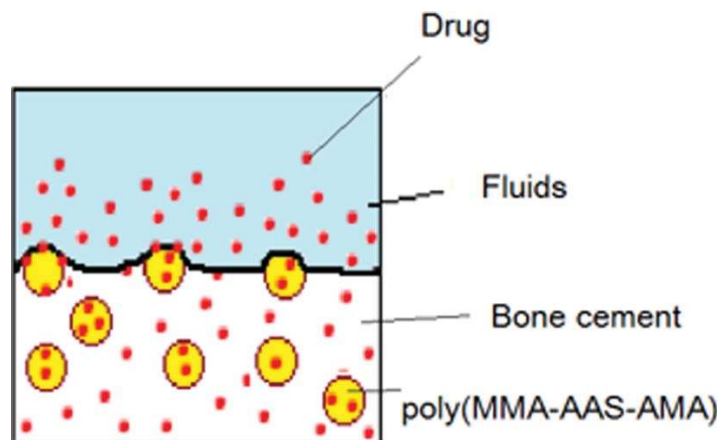


Figure 6. Schematic of drug release in bone cement, where the dots represent the drugs, the small circles represent the cross-linked poly (MMA-AAS-AMA) particles, below the thick black line represents bone cement, and above the line represents the solution/fluid.⁵⁰ © Yu Hsun Nien. Reproduced by permission of Yu Hsun Nien. Permission to reuse must be obtained from the rightsholder.

properties of the cement. This was achieved by introducing cross-linked PMMA-acrylic acid sodium salt particles into the bone cement, which increases the hydrophilicity of the cement for the easy passage of fluids.⁵⁰

Shi et al. reported the application of PMMA as a matrix polymer in the preparation of a PMMA/gelatin/antibiotic construct. The polymer construct porosity and antibiotic (colistin) release were controlled by the ratio of the gelatin microparticles (GMPs) present in the PMMA matrix. The construct was capable of releasing the antibiotics for 10 or 14 days, with an average release of $10 \mu\text{g/mL}$ in a day, and the degradation of the construct was tested using phosphate-buffered saline-collagenase, while the porosity was found to be unchanged for the period of 12 weeks.¹² The porosity remained unchanged due to the good mechanical properties of PMMA to maintain the porosity, impacted by a certain ratio of the gelatin microparticles. The PMMA construct was found to have the mechanical ability to work sufficiently for the correction and repair of craniofacial contour deformities in clinical applications and animal models.

More scientific investigations are needed to improve or explore the mechanical capabilities of other biocompatible polymers, such as poly (2-hydroxyethyl methacrylate) and poly (vinyl pyrrolidone) as polymer matrices for drug delivery and cranioplastic applications.

Another study of the drug delivery system was reported by Langer et al., which also involves the use of MMA and sulfopropyl methacrylate (SPMA) monomers to produce copolymer nanoparticles that are used for the topical ophthalmic application of the muscarinic agonists arecaidine propargyl ester (APE) and (S)-(+)-aceclidine in rabbits, compared with conventional eye drops. This work proved the use of nanoparticles as carriers for the muscarinic agonists arecaidine propargyl ester and (S)-(+)-aceclidine, as they did not show any irritating effects in the rabbit eye.⁵³ Downes improved the drug release from PMMA bone cement using the method of polymer to monomer ratio. An increase in the polymer-to-monomer ratio increases the drug (antibiotic) release due to high porosity as a result of more polymer (PMMA) in the formulation than monomer (MMA), and vice-versa. It was found that the optimum release was achieved if the crystalline

formulation of the drug was used.⁵⁴ This work was among the earlier research studies that opened up a gateway for the potential exploitations of PMMA as a drug release polymer matrix.

PMMA was also used in the preparation of a hydroxyapatite-PMMA (HA-PMMA) composite. The composite was used successfully as an implant material for cranioplasty due to the osteoconductivity of HA, strength, and ease of handling of the PMMA, as reported by Itokawa et al. They carried out *in vivo* investigations of the new HA-PMMA composite by implanting the material in the skulls of full-grown beagles to verify its osteoconductivity and biocompatibility. The extirpated specimens from the sacrificed beagles were examined using an optical microscope and micro-computed tomography. These results demonstrated that mixing HA and PMMA does not interfere with the osteoconductivity of HA; hence, the HA-PMMA composite was found to be a promising candidate for cranial bone implants.⁵⁵

Thomson et al. reported the biocompatibility of four different commercial PMMA bone cements both *in vitro* and *in vivo*, where they studied the cell integrity and inflammation. Based on the results, the *in vitro* studies showed the inflammatory potential of the four bone cements due to cell damage. While *in vivo*, all four bone cements showed compatibility in the mouse knee when evaluated over the period of 2–52 weeks.⁵⁶

3.2 Applications in Molecular Separations

Molecular separations in chemistry can be achieved in an advanced way by the use of chromatographic techniques, which involve the use of a stationary phase (inert solid support) and mobile phase (usually solvent or mixture of solvents). The common solid supports used are inorganic materials such as silica and alumina, but they have low patronage due to their disadvantages in separating some organic molecules, and limited modifications for maximum efficiency. Therefore, both natural and synthetic polymers have recently been used to replace the inorganic materials. PMMA is the most promising synthetic polymer for applications in molecular separation due to its low cost, compatibility, ease of modification, and processability.

Tai et al. synthesized a magnetic porous microsphere of Fe_3O_4 -PMMA-co-divinyl benzene via modified suspension polymerization. The magnetic porous microsphere was used for the removal of phenol from the aqueous solution due to its large surface area and stability. The adsorption method is the principle behind the removal of the phenol because of the high efficiency of the adsorption of the magnetic porous microspheres. The product has many promising potential applications in many other fields, such as enzyme immobilization and biological separation.⁵⁷

Shah et al. reported that the surface modification of PMMA improved the adsorption of wall coating polymers for microchip electrophoresis. Several methods for the chemical modification of the PMMA surface were presented for the application of wall coating polymers to suppress the electroosmotic flow and reduce analyte wall interactions. The electrophoretic separation of ssDNA was successfully carried out in microchips produced using these methods. It was found that 10 min of UV irradiation to surfaces coated with hydroxypropylmethyl-cellulose yielded a better separation performance.⁷

Qi and Lee modified polypropylene hollow fibers with PMMA and produced a desired capillary electrophoresis column. The unsaturated functional groups in the polypropylene interacted with borane-dimethylsulfide; then the PMMA was grafted on the

surface to produce a stable coating that reduced the electroosmotic flow and adsorption of the biomolecules. The PMMA modified polypropylene exhibited a lower electroosmotic flow than the fused silica and polypropylene; therefore, it had the highest efficiency of separation.⁵⁸

The use of PMMA that boosts the efficiency of magnetic porous microspheres for the removal of selected organic compounds (molecules) from an aqueous media is a breakthrough not limited to laboratory applications, it also has industrial ones. Phenol and its derivatives have wide industrial applications; therefore, they are found in industrial wastewater. As such, PMMA magnetic porous microspheres could play an important role in industrial wastewater treatment for the removal of toxic and carcinogenic compounds like phenol and its derivatives. Considering the efficiency of magnetic porous microspheres, there is a need to explore more PMMA magnetic porous microspheres for the removal of dyes from the effluents of the coloration industries.

3.3 Optical Applications

Optical science is relevant and studied in many disciplines, including engineering, medicine, pure science, and astronomy. Practical applications are found in lenses, microscopes, lasers, fibers, and polymers, to name a few. The optical activity of any material is the result exhibited by that material when interacting with light and the refractive index is the measure of that activity. The optical applications of PMMA are due to its refractive index, good resistance to UV light, chemical durability, and good mechanical properties.

Organic polymers are usually cheap, lightweight, and easily processed substrates, and are therefore good for immobilizing semiconductors for heterogeneous photocatalytic applications. Cámara et al. revealed the investigation of eleven synthetic polymers susceptible to coating with TiO₂ for exposure to solar radiation, with and without the TiO₂ layer, for 150 days to study the weathering.

They observed that only the PMMA retains good optical and mechanical properties of the titania after natural weathering. Therefore, PMMA is the best candidate for the immobilization of TiO₂ for photocatalytic treatment applications.⁶⁰

Organic third-order nonlinear optical (NLO) materials, mostly dyes, have been given much attention because of their potential applications in the optical fields. Miao et al. investigated the third-order NLO properties of the spin-coating PMMA films containing 3,7-bis (diethylamino) phenoxazin-5-ium chloride as a dopant. From the results of the evaluations using Z-scan technology at 532 nm, it was found that the third-order NLO reverse saturation absorption changes to saturated absorption due to the molecular aggregation of the phenoxazinium dye. Conclusively, the absorption coefficient and nonlinear refractive index of the phenoxazinium dye are good, making it a candidate molecule for third-order nonlinear materials.⁶¹

Yamano and Kozuka published work where a hybrid thin film of PMMA and perhydropolysilazane-derived silica (PHPS) doped with spiropyran (SP) was prepared by spin coating. The photochromic properties of the spiropyran molecules in the as-prepared films in the form of merocyanine (MC) and sipro (SP) were investigated. The spiropyran/(SP+PHPS+PMMA) mass ratio was fixed at a high value of 0.2, and the PMMA/(PMMA+PHPS) mass ratio was varied. The degree of conversion from MC to SP upon visible light illumination was enhanced for the 0 to 0.2 fixed values, while at high values of $r = 0.4$ and above the spiropyran molecules underwent photodegradation. The role of

the PMMA in the film is to improve the chemical durability of the film and enhance the conversion of the MC to SP during visible light illumination.⁶²

Martinez et al. produced a fabricated carbon nanotube-doped PMMA composite using an MMA monomer without solvent for application in nonlinear photonic devices. In the process they also used dopants such as diphenyl sulphide (DPS) and benzyl benzoate, and the DPS was found to play a role as a dispersion stabilizer that can improve the device performance. The PMMA was used in the composite preparation to solve the problem of poor flexibility of the CNT, which can hinder the application of the CNT despite of its nonlinear optical properties.⁶³

Andreas Hinsch examined the fluorescence and photostability of organic dyes incorporated in $\text{Al}_2\text{O}_3\text{-SiO}_2$ and $\text{TiO}_2\text{-SiO}_2$ (ormocer) with the aim of combining the fluorescence properties of the dyes in the PMMA for a rigid ormocer matrix. The PMMA also increases the UV stability of the dyes with its ratio increase. The thin fluorescent coatings with good optical qualities allow the preparation of fluorescent light concentrators with a high concentration ratio for solar applications.⁶⁴

3.4 Applications in Polymer Conductivity

Most polymers are electrical insulators. However, conducting polymers can be prepared using an insulating polymer and electrically conductive fillers called dopants. The electrical properties of PMMA doped with conducting materials under various experimental conditions including photo-induced changes, has been studied.

Electrically conductive materials, such as carbon nanotubes, carbon black, graphite, and metallic powder, when added above the critical concentration into an insulating polymer have been good methods of producing conductive composites. Coelho et al. synthesized MWCNT grafted PMMA nanocomposites via the *in situ* polymerization of different dispersed concentrations of MWCNT in an MMA monomer in the presence of a 2,2-bis(isobutyro nitriles) (AIBN) initiator. The results illustrate an increase in the volumetric electrical conductivity of the matrix, up to 10 orders of magnitude, making the dielectric PMMA into a semiconductor.⁶⁵

PMMA has been used recently as an insulating layer for organic memory devices, and found to be suitable for the task. Haik et al. fabricated and characterized novel organic memory devices made of Au-Pt-Ag nanoparticle PMMAs and poly-vinyl-alcohol/poly acrylamide co-acrylic acid with glycerol ionic liquids (PVA-PAA-glycerol). The PMMA was used as an organic insulator, while the PVA-PAA-glycerol was a semiconducting polymer. Aluminum electrodes were used as bottom and top electrodes for the fabricated devices. Finally, organic memory devices were prepared based on the Au-Pt-Ag nanoparticles as charge storage elements. Herein, PMMA was used as the organic insulator.⁶⁶

Liu et al. communicated the conductivity of disk-like PMMA/carbon black composites by mapping different areas on the disk for the analysis. The disk-like composite was divided into several small pieces, and the electrical conductivity was investigated for each piece. They found that the electrical conductivity of the static (nonshared) sample at the center of the disk was independent of the radius. For the shared sample, the conductivity decreased from the center to the outer rim of the disk-like sample.⁶⁷

Biswas et al. determined the surface electrical conductivity of a PMMA thin film prepared in a chloroform solvent by dissolving and stirring before casting the film. The purpose of the study was to investigate the conductivity switching of the prepared film under certain experimental conditions. Conductivity switching was observed with a rise in

temperature and the illumination of light, while no conductivity was found in the dark and at room temperature. The conductivity of the PMMA may be due to the small rise in the temperature and the illumination of the light that resulted in the reorganization/reorientation of the PMMA structure (molecular conformation) that causes the redistribution of charges in the molecule.⁶⁸

The reversibility in terms of conductivity exhibited by the PMMA thin film after the removal of the voltage and the illuminated light has illustrated its suitability for applications in electronic devices. Sometimes, the electric conductivity of the polymer composite is strongly influenced by the structure of the particle network of the conductive filler. Therefore, to obtain a material with desirable electrical properties, we need to change the filler/polymer network by deformation in the molten state.

Krückel et al. reported the investigation of a PMMA composite filled with carbon black and carbon fiber. When exposed to oscillatory shear deformations, the electrical conductivity of the materials was measured. Based on the results, the carbon black composite with a lower percolation threshold had a higher conductivity when compared to the carbon fiber composite. It was concluded that conductivity measurements are sensitive tools to follow the changes in a particle network, even if the rheological quantities remain unaffected.⁴⁴

Hwang et al. reported the fabrication of a composite of piezoelectric α -helical poly (α -amino acids), poly (γ -benzyl α , L- glutamate) (PBLG), and a matrix polymer of PMMA. The fabrication was done using a new method of contact charging of the PBLG-MMA solution mixture in the presence of an initiator, followed by curing the matrix in a designed mold chamber in between two electrodes. This allows strict control of the PBLG-PMMA composition through minimizing MMA evaporation. The results show that all the PBLGs were pooled in the direction normal to the disk surface, and are responsible for the piezoelectricity of the composite.⁶⁹

3.5 Applications in Polymer Electrolytes

A polymer electrolyte membrane for battery application must play the following roles: must enable positive ion transport such as Li^+ between the electrodes, must block the electron transport, and must be rigid to prevent direct contact between the electrodes.⁷⁰ The application of PMMA in the polymer electrolyte was due to the amorphous nature for porosity's sake and the mechanical strength it has for the provision of the rigidity to the polymer electrolyte membrane.

Recently, the use of PMMAs in composing polymer electrolytes was focused on by researchers. PMMA was directly responsible for the porosity, strength, and thermal stability of the electrolytes. Flora et al. composed polymer electrolytes of poly (acrylonitrile)/poly (methyl methacrylate)/lithium perchlorate (PAN/PMMA/ LiClO_4) with different plasticizers such as ethylene carbonate and poly carbonate (EC and PC), among others. The results showed a maximum ionic conductivity of the order of $0.661 \times 10^{-4} \text{ Scm}^{-1}$ for EC-based electrolytes at room temperature. The thermal stability and conductivity confirmed that the PAN/PMMA/ LiClO_4 -EC polymer electrolyte was found to be suitable for applications in lithium ion batteries.⁷¹

The credibility of PMMA in the hybrid of poly (vinylidene fluoride) (PVDF) was attributed to its amorphous nature and excellent compatibility with PVDF. The role of PMMA in the base gel polymer electrolyte of PVDF is to increase ionic conductivity by neutralizing the crystallinity of the PVDF. Zhou et al. reported the use of a PVDF/PMMA electrospun matrix as the base gel polymer with and without *in situ* TiO_2 in the

preparation of gel polymer electrolytes by activating in a 1M LiClO₄-EC/PC liquid electrolyte. The results show that the addition of TiO₂ improved the ionic conductivity at room temperature to $3.9 \times 10^{-3} \text{ Scm}^{-1}$, with a chemical stability of up to 5.1V. They concluded that the gel polymer electrolyte prepared with *in situ* TiO₂ was very suitable for applications in Li-ion batteries.⁷²

Kufian et al. investigated the application of PMMAs in polymer electrolytes and reported the conductivity of a homogenous mixture of different amounts of lithium bis (oxolato) borate (LiBOB) prepared with equal weights of ethylene carbonate (EC) and propylene carbonate (PC). The mixture with the highest room temperature conductivity was further used to produce a gel polymer electrolyte with 10 to 50 wt.% PMMA by casting and heating between 70 and 80°C for 20 min. The result of further investigation of the PMMA gel polymer electrolyte (GPE) showed that the GPE sample with the 20 wt.% PMMA content was found to be an ionic conductor with a lithium ion transference number of 0.46 at room temperature. Therefore, the PMMA-LiBOB GPE was suitable for use in lithium batteries, because a linear sweep examination indicated that the gel electrolyte is stable until 4.7 V.⁷³

Another application of PMMA in a gel polymer electrolyte for lithium batteries was reported by Ali et al. They produced a gel polymer electrolyte (GPE) consisting of 30% PMMA-grafted natural rubber (MG30), lithium triflate (LiTf), and ethylene carbonate (EC) by dissolving the mixture in THF, followed by drying the residual solvent to obtain the GPE. The AC impedance technique was used to optimize the conductivity of the gel samples at room temperature in the frequency range of 0.1 KHz and 1.0 MHz. The gel composition containing 35 wt.% of LiTf was found to exhibit the highest ionic conductivity.⁷⁴ The role of PMMA in this gel polymer electrolyte was to make the cell membrane electrolyte rigid for electrochemical stability, while ethylene carbonate (EC), acting as a polar solvent, increased the ionic conductivity without disturbing the complexation between the polymer and the lithium salt.

Ramesh et al., in the Journal of Non-Crystalline Solids, communicated their achievement in evaluating the effect of an ionic liquid of 1-butyl-3-methylimidazolium bis (trifluoro methylsulfonylimide) BmImTFSI onto a PMMA-PVC gel polymer blending electrolytes. It was found that the ionic conductivity of $(8.08 \pm 0.01) \times 10^{-4} \text{ Scm}^{-1}$ was achieved at 80°C by the addition of 60 wt.% of BmImTFSI.⁷⁵

The copolymer of a PMMA-clay nanocomposite was also used in the preparation of gel polymer electrolytes as the host matrix. In the work published by Deka and Kumar, the effect of organically modified montmorillonite (MMT) clay on PMMA-based electrolytes was investigated. It was observed that the ionic conductivity of the nanocomposite gel polymer electrolyte increased with an increase in the clay loading to a maximum value of $1.3 \times 10^{-3} \text{ S/cm}$ at room temperature.⁷⁶ Not only nanoclay, even silica fume in nanoscale was incorporated into the PMMA host matrix in the presence of a conductive ion filler to produce gel polymer electrolytes. The thermal stability of the polymer gel electrolyte shows an improvement with the addition of fumed silica up to the temperature of 125°C.⁷⁷

Due to its wider applications in the field of battery electrolytes, PMMA was used not only in lithium battery electrolytes, but also in magnesium battery electrolytes to produce a solid-state rechargeable magnesium cell. Girish and Munichandraiah reported the preparation of a gel polymer electrolyte of PMMA, propylene carbonate (PC), ethylene carbonate (EC), and magnesium triflate (MgTr) conductive filler. After the GPE was investigated for ionic conductivity, transport number, and

stability, the GPE with an optimum conductivity of $(4.2 \pm 0.05) \times 10^{-4} \text{ S cm}^{-1}$ was used to produce a solid-state rechargeable magnesium cell.⁷⁸ Agnihotry et al. communicated the preparation of gel electrolytes based on PMMA and a solution of lithium salt $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ in polypropylene carbonate (PC) and a mixture of polypropylene carbonate (PC) + ethylene carbonate (EC). A comparison of the liquid electrolytes with the salts of LiCF_3SO_3 and $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ was done. The excellent conductivity exhibited by the gels, and their moderate variations in the operational temperature range of electrochromic devices (ECDs), make them potential solid state ECDs for window applications.⁷⁹

Several appearances of PMMA in the literature report on the preparation of gel polymer electrolytes for both lithium and magnesium batteries, showing potentials in industrial applications. This was attributed to its good mechanical strength, low cost, amorphous nature, compatibility, thermal stability, and a refractive index value which is almost the same as that of glass. Researchers and industrialists should pay attention to this promising polymer for more explorations of its industrial applications. However, it shows a decrease in the conductivity of gel polymer electrolytes when added beyond 40% in ratio to the conductive fillers.

3.6 Viscosity Applications

The viscosity of a fluid is the measure of its resistance to flow under applied shear stress. Polymeric melts or solutions are non-Newtonian fluids; therefore, the share stress is not directly proportional to the share rate. However, the polymers have intrinsic viscosity, which is a measure of their ability to increase the viscosity of another fluid. For this reason, a polymer of high molecular weight can alter or modify the viscosity of low molecular weight polymers. Due to its compatibility and easy processability, PMMA can be grafted to natural polymers to obtain high molecular weight copolymers of high intrinsic viscosity, with a potential application as a viscosifier. Mishra and Sen reported this scenario in their work grafting guar-gum with PMMA using a microwave initiated method. The correlation between the percentage of grafting and the intrinsic viscosity of the product was evaluated, and the results show that it can be used as a superior viscosifier to guar-gum alone.¹³

3.7 Applications in Pneumatic Actuation

Pneumatic actuators are devices that consist of a piston, cylinder, and valves or ports. The piston is usually sealed to keep air in the upper or down portion of the cylinder, depending on the design of the pneumatic actuator. Their purpose is to control or regulate the flow of fluids in a tubular channel perpendicular to the pneumatic actuator. However, in micro-scale, pressurized air can be switched to the pneumatic actuators by solenoid valves through electronic control for successful fluid pumping. It was recently revealed that PMMA happens to be a potential candidate used in a composite for the construction of pneumatic actuators for micropumps.

Douglas et al. presented an electrostatic microvalve for the pneumatic control of microfluidic devices. The valve was manufactured by depositing a thin chrome layer on PMMA and copper foil as a flexible membrane. The results showed that the valve had a low closing voltage and high flow rate. They concluded that the valve was a unique one, because it was easy, inexpensive to manufacture, and had the ability to integrate with complex polymer fluidic systems.⁸⁰

Tan et al. reported the novel technique of bonding PMMA and poly (dimethylsiloxane) (PDMS) in a multi-layer polymeric microfluidic device. The lamination of a thin layer of dry adhesive was done on the PMMA, followed by spin coating and curing of the PDMS layer. They concluded that the product was a good material to use in peristaltic micropumps for the construction of pneumatic actuators. The role of PMMA in the laminated layer is to strongly bridge the adhesive/PDMS to form a hybrid membrane, and improve the gas impermeability of the adhesive layer.⁸¹

3.8 Sensor Applications

Chemical sensing technology is basically the chemical recognition of analytes and transduction of the analytical signal. A chemical sensor should also be capable of responding to concentration changes of the analytes in both liquid and gaseous media. A material to be used for chemical sensing must have electrical properties, a sensitivity to charge transfer, the chemical doping effect of various molecules, and be capable of immobilization. Therefore, taking into consideration the qualities of the materials for the chemical sensors, a CNT and PMMA composite or modified CNT would be a very good material for chemical sensors. However, recent developments have revealed the use of polymeric materials only in temperature sensing for applications in anti-counterfeit labels.

Haibao et al. reported the investigation of the heating/ethanol response of a commercial PMMA that was programmed with a gradient pre-strain/stress field, and used the features identified to demonstrate the potential applications of PMMA in temperature sensors. They used two types of PMMA molds: a Singaporean coin and a particular design mold with variable protrusive features on top. The basis of their analysis was compression to introduce a gradient pre-strain/stress field, and monitoring overheating temperatures with a water mark that appeared only upon heating to a particular temperature. They concluded that since the shape memory effect (SME) is an intrinsic feature of most polymers, other conventional polymers may be used in such applications.⁸²

Polydiacetylene (PDA) is one of the conjugated polymers with a visible color transformation induced by varied external stimuli such as heat, ionic strength, and interactions with biological and chemical molecules. PMMA was found to be compatible by spin-coating with PDA for sensing applications. Parambath et al. presented a new PDA film prepared through spin-coating upon PMMA, and they investigated the photophysical properties and sensing applications of the new PDA film. Considering the results obtained, they concluded that the study presents a new chromatic PDA film system giving rise to distinct chromatic properties and molecular sensing capabilities.⁸³

Following the potential applications of the surface plasmon resonance (SPR) based sensor, researchers employed an extensive research effort in this area during the past two decades. Mishra et al. fabricated a probe by coating an unclad portion of an optical fiber first with copper, then by sensing an over-layer of reduced graphene oxide (rGO), PMMA, and PMMA/rGO nanocomposites. Each sensor operated in the wavelength interrogation scheme and was based on the reactivity of ammonia gas with the sensing over-layer. The results showed that the PMMA/rGO nanocomposite over-layer coating was found to be more sensitive than the rGO and PMMA over-layer coatings with the ammonia gas. Due to its high sensitivity, wider operating range, reusability, and reproducibility, the PMMA/rGO nanocomposite over-layer was found to be suitable for practical applications.⁸⁴

Michael et al. presented the use of a single crystal resonator of lead magnesium niobate-lead titanate (PMN-PT) and PMMA to produce a thin film useful for detecting chemicals such as acetone, methanol, and isopropyl alcohol. From the results obtained, the PMMA coated PMN-PT showed high sensitivity and significant selectivity when compared to a PMMA coated quartz crystal microbalance (QCM).⁸⁵

Hargsoon et al. designed and developed a passive wireless sensor for the detection of bio-hazardous materials. They used a composite polymer thin film of functionalized carbon nanotubes (CNTs) and PMMA. The results showed a large differential phase shift, which can be utilized for real time monitoring.⁸⁶

PMMA without CNT or any other electrical conductive material can be used to spin into a thin film for the chemical sensing of benzene molecules. Capan et al. reported the processing of PMMAs of different molecular weights into thin films on a solid support. The film thickness increased with the molecular weight, and the sensitivity of the benzene detection was dependent on the molecular weight. In addition, the results revealed that the film had a fast and reversible response to the benzene vapor. Therefore, the PMMA film has potential applications in vapor sensing membranes.⁸⁷

Isha et al. communicated the sensing of vanadium (V) metal, or its derivatives, using fatty hydroxamic acid (FHA) immobilized in PMMA. Based on the results, despite the interference of the Fe (III) ion in the V (V) determination, the results were as good as those obtained using the Inductively Coupled Plasma Atomic Emission Spectroscopy (ICP-AES) method.⁹

3.9. Solar Applications

In search of the production of a quasi-solid-state dye-sensitized solar cell (DSSC) using a high conductivity polymer gel electrolyte, a suitable polymeric material was needed to be a host matrix in the composite. Therefore, PMMA was found to be a good and compatible polymer for this purpose. This was attributed to its mechanical strength, compatibility, and optical clarity.

Shen et al. very recently, and for the first time, reported the synthesis of europium ion (Eu^{3+}) doped sodium gadolinium fluoride (NaGdF_4 : Eu) nanocrystals (NCs) using a hydrothermal method. The down conversion (DC) layer, consisting of PMMA doped with luminescent NaGdF_4 : Eu, was prepared and attached to the back of TiO_2 anodes for efficient DSSCs. The results of the evaluation of the effect of doped and undoped NaGdF_4 nanocrystal layers on the photovoltaic device with incident-photon-to-current efficiency (IPCE) as the parameter of comparison showed that the DSSC with a doped NaGdF_4 : Eu DC-PMMA layer improved the photoelectric conversion efficiency by 4.5%.⁸⁸

Yan et al. prepared a polymer gel electrolyte of PMMA, ethylene carbonate, 1, 2-propanediol carbonate, dimethyl carbonate, and sodium iodide/iodine as a source of I^-/I_3^- . The polymer electrolyte PMMA-EC/PC/DMC- NaI/I_2 was found to have a high ionic conductivity of 6.89 mScm^{-1} , and was used to fabricate a quasi-solid-state dye-sensitized solar cell (DSSC) with good long term stability and a light-to-electrical energy conversion efficiency of 4.78%.⁸⁹

Hammam et al. produced fluorescent PMMA film embedded with coumarin dyestuff (MACROLEX Fluorescent Red G) commercial dye using a flow-spin coating technique. The film with a dye concentration corresponding to the maximum intensity was used, and its emission was optimized to match the absorption bands of chlorophyll (650–680 nm) for greenhouse applications. Due to its excellent weathering durability, it can be used in growing rooms in which plants are grown for commercial purposes.⁹⁰

For the sake of advancement in technology, Schissel et al. reported the preparation of metallized flexible polymeric reflectors that are much lighter and less expensive than the conventional glass/metal mirrors. This product can replace conventional glass/metal mirrors in solar thermal concentrators, such as heliostats and parabolic dishes, with little modification for long-term weathering upon corrosion.⁹¹

3.10. Applications in Nanotechnology

The existence of interactions between polymers and nanomaterial opened up a serious breakthrough in the field of nanotechnology. Polymer nanocomposites were the products of this interaction that offered a substantial improvement in material properties with a very small amount of nanoparticles. The material properties that could be improved are mechanical strength, solubility, electrical conductivity, optical properties, scratch resistance, thermal stability, and flame retardation, to mention a few.

Due to the wider application of nanocomposites in the field of nanotechnology, many researchers focused their attention on nanocomposites, their fabrication, and applications. Perween et al. reported the use of PMMA and graphite to fabricate plastic chip electrodes (PCEs) via a simple solution casting method. This characterization was made using microscopy (SEM and AFM) as well as thermal properties (TGA), and mechanical and electrical properties. The fabricated electrode was economically inexpensive, multipurpose, and dispensable for various applications.⁹²

An advancement in the method (approach) of preparing nanocomposites using nanoparticles was found to be in the combination of both covalent and noncovalent interactions. Wang et al. investigated the modification effect of SiO₂ nanoparticles in the composite of SiO₂/PMMA. The composite was fabricated by a two-step process: the noncovalent modification of the SiO₂ nanoparticles with tetraoctylammonium bromide to assist the dispersion of the nanoparticles in the solvent, and the covalent process by radical suspension polymerization with the MMA monomer to yield the silicon oxide/PMMA nanocomposite. The aim was to improve the mechanical properties of PMMA using SiO₂ for wider applications. They concluded that the tensile strength and flexural strength were enhanced by as much as 80.6% and 127.3% when compared to those of pure PMMA, respectively. The two-step approach was found to be simple, efficient, and cost-effective.⁹³

On the other hand, the surface functionalization of nanoparticles via the grafting of a polymer plays an important role in the design of both organic and inorganic nanocomposite materials. Atom Transfer Radical Polymerization (ATRP) was the latest method used for the grafting because of its better control over molecular weight and its low polydispersity. The surface-initiated ATRP technique is widely used for the grafting of homopolymers, diblock copolymers, graft copolymers, star polymers, and branched polymers from nanoparticles, nanotubes, nanowires, and nanoclays.⁹⁴

PMMA was grafted from carbon nanotubes (CNTs) to improve the solubility and processability of the CNTs. This was found to be of importance considering the outstanding low density, mechanical, electrical, and thermal properties of the CNTs, with limited solubility because of the high tendency of aggregation due to the π - π bond interactions.

Wang et al., in their work enhancing the mechanical properties of poly (styrene-co-acrylonitriles) with PMMA-grafted MWCNT, synthesized a PMMA-*g*-MWCNTs copolymer by Atom Transfer Radical Polymerization (ATRP). According to the results, the grafting was successful.¹⁴

Cui et al. reported the grafting of PMMA onto single walled carbon nanotubes via *in situ* radical polymerization. A free radical initiator with difunctional carboxyl groups was

covalently attached to SWCNT-OHs by esterification, then the free radical initiator was attached, followed by polymerization using an MMA monomer.⁹⁵

Another grafting of PMMAs onto MWCTs was reported by Baskaran et al., where the PMMA was synthesized to have hydroxyl-terminated groups and grafted onto the acid chloride MWCNTs using an esterification reaction.⁴⁵

Sung et al. prepared a PMMA/MWCNT nanocomposite by the effective dispersion of MWCNT via the *in situ* bulk polymerization of MMA. Nanofibrous membranes of the composite were also produced by electrospinning. The purpose of the study was to find any variations in the electrical conductivities of the two forms of the composite (bulk PMMA/MWCNT and electrospun PMMA/MWCNT). The results demonstrated that the bulk nanocomposite conductivity was between 10^{-4} and 10^{-2} Scm⁻¹, while the electrospun film conductivity was reduced to 10^{-10} Scm⁻¹. Conclusively, the electrospinning process was not very efficient in enhancing the electrical conductivity of the polymer matrix.⁹⁶

References

1. Malcom, P.S. *Polymer Chemistry: An Introduction*. 3rd ed.; Oxford University Press: NY, pp 167–176, 256–276.
2. Henri, L. *Thermohygroelastic Properties of Polymethylmethacrylate*; 2007, Netherlands. pp. 11–13.
3. Demir, M. M.; Memesa, M.; Castignolles, P.; Wegner, G. "PMMA/zinc oxide Nanocomposites prepared by in-situ bulk polymerization", *Macromolecular Rapid Communications*. **2006**, 27 (10), 763–770.
4. Hashim, H.; Adam, N. I.; Zaki, N.H.M.; Mahmud, Z.S.; Said, C.M.S.; Yahya, M.Z.A.; Ali, A.M.M. "Natural rubber-grafted with 30% Poly (methylmethacrylate) characterization for application in lithium polymer battery", Conference on Science and Social Research 2010 (CSSR 2010), Kuala Lumpur, Malaysia, 485–488.
5. Henry, A.C.; Tutt, T.J.; Galloway, M.; Davidson, Y.Y.; McWhorter, C.S.; Soper, S.A.; McCauley, R.L. "Surface modification of poly(methyl methacrylate) used in the fabrication of micro-analytical devices", *Analytical Chemistry* **2000**, 72(21), 5331–5337.
6. Lee, L.H.; Chen, W.C. High-refractive-index thin films prepared from trialkoxysilane-capped poly (methyl methacrylate), Titania materials. *Chem. Mater.* **2001**, 15, 1137–1142.
7. Shah, J. J.; Geist, J.; Locascio, L. E.; Gaitan, M.; Rao, M. V.; Vreeland, W. N. "Surface modification of poly(methyl methacrylate) for improved adsorption of wall coating polymers for microchip electrophoresis", *Electrophoresis* **2006**, 27(19), 3788–3796.
8. Adhikari, B.; Majumdar, S. "Polymers in sensor applications", *Progress in Polymer Science* **2004**, 29(7), 699–766.
9. Isha, A.; Yusof, N.A.; Ahmad, M.; Suhendra, D.; Yunus, W.M.Z.W.; Zainal, Z. "A chemical sensor for trace V(V) ion determination based on fatty hydroxamic acid immobilized in polymethylmethacrylate", *Sensors and Actuators B: Chemical*. **2006**, 114(1), 344–349.
10. Kost, J.; Langer, R. "Responsive polymeric delivery systems", *Advanced Drug Delivery Reviews*. **2012**, 64, 327–341.
11. Beruto, D.T.; Botter, R.; Fini, M. "The effect of water in inorganic microsponges of calcium phosphates on the porosity and permeability of composites made with polymethylmethacrylate", *Biomaterials*. **2002**, 23(12), 2509–2517.
12. Shi, M.; Kretlow, J.D.; Spicer, P.P.; Tabata, Y.; Demian, N.; Wong, M.E.; Kasper, F.K.; Mikos, A.G. "Antibiotic-releasing porous polymethylmethacrylate/gelatin/antibiotic constructs for craniofacial tissue engineering", *Journal of Controlled Release* **2011**, 152(1), 196–205.
13. Mishra, S.; Sen, G. "Microwave initiated synthesis of polymethylmethacrylate grafted guar (GG-g-PMMA), characterizations and applications", *International Journal of Biological Macromolecules*. **2011**, 48(4), 688–694.

14. Wang, M.; Pramoda, K.P.; Hong, S. "Enhancement of the mechanical properties of poly (styrene- co -acrylonitrile) with poly (methyl methacrylate) -grafted multiwalled carbon nanotubes", *Polymer* **2005**, 46, 11510–11516.
15. George, O. *Principles of Polymerization*, 4th ed.; Wiley: NJ, 2004, pp 198–235.
16. Wang, J.-S.; Matyjaszewski, K. "Controlled/"living" radical polymerization. atom transfer radical polymerization in the presence of transition-metal complexes", *Journal of the American Chemical Society*. **1995**, 117(20), 5614–5615.
17. Grimaud, T.; Matyjaszewski, K. "Controlled/"living" radical polymerization of methyl methacrylate by atom transfer radical polymerization", *Macromolecules*. **1997**, 30(7), 2216–2218.
18. Charles, A.H.; Edward, M.P. *Plastics Materials and Processes, in Concise Encyclopedia*; Wiley: NJ, 2003, pp. 42–44.
19. Van Krevelen, D.W.; Nijenhuis, K. T. *Properties of Polymers*; Elsevier: Amsterdam, 2000, pp. 106, 322.
20. Charles, A.H. *Handbook of Plastics Processes*; Wiley: NJ, 2006, pp. 1–7.
21. Chang, L.; Woo, E.M. "Tacticity effects on glass transition and phase behavior in binary blends of poly (methyl methacrylate)s of three different configurations", *Polymer Chemistry* **2010**, 1(2), 198–202.
22. Chen Chuanfu, R.C.; Xi Fu "Stereoregularity of poly (methyl methacrylate) obtained with chiral anionic initiator", *Chinese Journal of Polymer Science*. **1995**, 13(1), 91–96.
23. Ishitake, K.; Satoh, K.; Kamigaito, M.; Okamoto, Y. "From-syndiotactic-to-isotactic stereogradient methacrylic polymers by RAFT copolymerization of methacrylic acid and its bulky esters", *Polymer Chemistry*. **2012**, 3(7), 1750–1757.
24. Li, L.; Li, B.; Chen, J.; Zhou, D.; Xue, G.; Liu, X. "Glass transition and structural relaxation of nano-particle aggregates of atactic poly(methyl methacrylate) formed in microemulsions", *Polymer* **2004**, 45(8), 2813–2816.
25. Mark, J. E. *Physical Properties of Polymers Handbook*, 2nd ed.; Springer: NY, 2007, pp. 3–8.
26. Semen, J.; Lando, J. B. "The acid hydrolysis of isotactic and syndiotactic poly (methyl methacrylate)", *Macromolecules*. **1969**, 2(6), 570–575.
27. www.azom.com/article.aspx. Polymethylmethacrylate - Acrylic - PMMA General purpose. Cited June 4, 2014–Nov. 22, 2014.
28. Kutz, M. *Handbook of Materials Selection*; 2002, Wiley: NJ, p. 341.
29. Zeng, W.R.; Li, S.F.; Chow, W.K. "Preliminary studies on burning behavior of polymethylmethacrylate (PMMA)", *Journal of Fire Sciences* **2002**, 20(4), 297–317.
30. Zhi, C.Y.; Bando, Y.; Wang, W. L.; Tang, C.C.; Kuwahara, H.; Golberg, D. "Mechanical and thermal properties of polymethyl methacrylate-bn nanotube composites", *Journal of Nanomaterials*. **2008**, 1–5. doi:10.1155/2008/642036.
31. Ban, M.; Yuhara, T. "Chemical resistance of DLC thin film deposited PMMA substrates", *Surface and Coatings Technology*. **2009**, 203(17–18), 2587–2590.
32. Jian Li, Y.L.; Meng, X.; Liu, Y. "Durability of wood-PMMA composite", *Applied Mechanics and Materials*. **2010**, 26–28, 830–834.
33. Pan, X.; Mercadé-Prieto, R.; York, D.; Preece, J. A.; Zhang, Z. "Structure and mechanical properties of consumer-friendly PMMA microcapsules", *Industrial & Engineering Chemistry Research*. **2013**, 52(33), 11253–11265.
34. www.sevierlab.vet.cornell.edu/resources/Chemical-Resistance-Chart-Detail.pdf. Chemical Compatibility Guide: Chemical Resistance Chart Detail. cited **2015**, Feb 11.
35. www.solutions-in-plastics.info. PMMA Resistance to Chemical. cited **2015**, Feb 11.
36. Ueno, K.; Fukai, T.; Nagatsuka, T.; Yasuda, T.; Watanabe, M. "Solubility of poly(methyl methacrylate) in ionic liquids in relation to solvent parameters", *Langmuir* **2014**, 30(11), 3228–3235.
37. Evchuk, I.Y.; Musii, R.I.; Makitra, R.G.; Pristanskii, R. E. "Solubility of polymethyl methacrylate in organic solvents", *Russian Journal of Applied Chemistry* **2005**, 78(10), 1576–1580.
38. Zeng, W.R.; Li, S.F.; Chow, W.K. "Review on chemical reactions of burning poly(methyl methacrylate) PMMA", *Journal of Fire Sciences* **2002**, 20(5), 401–433.
39. Chin, I.J.; Choi, H.W.; Lee, Y.; Won, H.Y.; Kim, Y.S. "Conversion of Poly (methyl methacrylate) to poly glutarimide", *Korea Polymer Journal* **1995**, 3(2), 76–81.

40. Bulmuş, V.; Ayhan, H.; Pişkin, E. "Modified PMMA monosize microbeads for glucose oxidase immobilization", *The Chemical Engineering Journal and the Biochemical Engineering Journal* **1997**, 65(1), 71–76.
41. Fixe, F.; Dufva, M.; Telleman, P.; Christensen, C.B. "Functionalization of poly(methyl methacrylate) (PMMA) as a substrate for DNA microarrays", *Nucleic Acids Research*. **2004**, 32(1), 1–8.
42. Amit, B.; James, W.R.; Paramita, R. *Polymer Grafting and Crosslinking*; Wiley: NJ. pp. 3–89, 2009.
43. Jin, Z.; Pramoda, K.P.; Xu, G.; Hong, S. "Dynamic mechanical behavior of melt-processed multi-walled carbon nanotube / $\pi\alpha\lambda\psi$ (methyl methacrylate) composites", **2001**, 337(3), 43–47.
44. Krücker, J.; Stary, Z.; Triebel, C.; Schubert, D. W.; Münstedt, H. "Conductivity of polymethylmethacrylate filled with carbon black or carbon fibres under oscillatory shear", *Polymer*, **2012**, 53(2), 395–402.
45. Baskaran, D.; Dunlap, J.R.; Mays, J.W.; Bratcher, M.S. "Grafting efficiency of hydroxy-terminated poly(methyl methacrylate) with multiwalled carbon nanotubes", *Macromolecular Rapid Communications*. **2005**, 26(6), 481–486.
46. Rani, P.; Mishra, S.; Sen, G. "Microwave based synthesis of polymethyl methacrylate grafted sodium alginate: Its application as flocculant", *Carbohydrate Polymers* **2013**, 91(2), 686–692.
47. Routray, C.R.; Biranchina, R.T.; Nayak, N. "Grafting of polymethyl methacrylate on to cellulose acetate in homogeneous medium using ceric (iv) ion as initiator", *Indian J. Chem. Technology* **2013**, 20, 202–209.
48. Harish, P.K.V.; Lakshman, K.; Shamala, T.R.; Tharanathan, R.N. "Biodegradation of chitosan-graft-polymethylmethacrylate films", *International Biodeterioration & Biodegradation*. **2005**, 56(2), 115–120.
49. Barroso, T.; Temtem, M.; Casimiro, T.; Aguiar-Ricardo, A. "Development of pH-responsive poly(methylmethacrylate-co-methacrylic acid) membranes using scCO₂ technology. Application to protein permeation", *The Journal of Supercritical Fluids* **2009**, 51(1), 57–66.
50. Nien, Y.H.; Lin, S.W.; Hsu, Y.N. "Preparation and characterization of acrylic bone cement with high drug release", *Materials Science & Engineering C* **2013**, 33(2), 974–978.
51. Zuhair, J.; Abdul Amer, J.K.A.; Sura Fahim, A. "Chitosan/PMMA bioblend for drugs release applications", *International Journal of Engineering and Technology* **2014**, 4(5), 318–324.
52. Zuber, M.; Tabasum, S.; Jamil, T.; Shahid, M.; Hussain, R.; Feras, K.S.; Bhatti, K.P. "Biocompatibility and microscopic evaluation of polyurethane–poly(methyl methacrylate)–titanium dioxide based composites for dental applications", *Journal of Applied Polymer Science* **2014**, 131(3), 1–9.
53. Langer, K.; Mutschler, E.; Lambrecht, G.; Mayer, D.; Troschau, G. "Methylmethacrylate sulfo-propylmethacrylate copolymer nanoparticles for drug delivery Part III : Evaluation as drug delivery system for ophthalmic applications", *International Journal of Pharmaceutics* **1997**, 158: 219–231.
54. Downes, S. "Methods for improving drug release from poly (methyl) methacrylate bone cement". *Clinical Materials* **1991**, 7, 227–231.
55. Itokawa, H.; Hiraide, T.; Moriya, M.; Fujimoto, M.; Nagashima, G.; Suzuki, R.; Fujimoto, T. "A 12 month in vivo study on the response of bone to a hydroxyapatite-polymethylmethacrylate cranioplasty composite", *Biomaterials* **2007**, 28(33), 4922–4927.
56. Thomson, L.A.; Law, F.C.; James, K.H.; Matthew, C.A.; Rushton, N. "Biocompatibility of particulate poly (methyl methacrylate) bone cements: A comparative study in vitro and in vivo", *Biomaterials* **1992**, 13(12), 811–818.
57. Tai, Y.; Wang, L.; Gao, J.; Amer, W.A.; Ding, W.; Yu, H. "Synthesis of Fe₃O₄@poly(methylmethacrylate-co-divinylbenzene) magnetic porous microspheres and their application in the separation of phenol from aqueous solutions", *Journal of Colloid and Interface Science*. **2011**, 360(2), 731–738.
58. Qi, S.; Lee, M.L. "Polypropylene hollow fibers modified with PMMA for capillary electrophoresis", *Journal of Microcolumn Separations*. **1998**, 10(7): 605–609.
59. www.dynachemplastic.en.made-in-china.com. PMMA Plastic Resin. cited 19th Feb., **2015**.

60. Camara, R.M.; Portela, R.; Gutiérrez-Martin, F.; Sánchez B. "Evaluation of several commercial polymers as support for TiO₂ in photocatalytic applications", *Global NEST Journal* **2014**, 6(3), 525–535.
61. Miao, J.-T.; Sun, R.; Yan, B.-L.; Wu, X.-Z.; Song, Y.-L.; Lu, J.-M.; Ge, J.-F. "Adjustable third-order nonlinear optical properties of the spin coating phenoxazinium-PMMA films", *Materials Chemistry and Physics* **2014**, 147(1–2), 232–237.
62. Yamano, A.; Kozuka, H. "Perhydropolysilazane-derived silica-polymethylmethacrylate hybrid thin films highly doped with spiropyran: Effects of polymethylmethacrylate on the hardness, chemical durability and photochromic properties", *Thin Solid Films*. **2011**, 519(6), 1772–1779.
63. Martinez, A.; Uchida, S.; Song, Y.-W.; Ishigure, T.; Yamashita, S. "Fabrication of Carbon nanotube-poly-methyl-methacrylate composites for nonlinear photonic devices", *Optics Express* **2008**, 16(15), 1581–1583.
64. Hinsch, A. "Organic fluorescent dyes in organically modified Al₂O₃-SiO₂ or TiO₂-SiO₂ coatings with varying polymethylmethacrylate content", *Journal of Non-crystalline Solids* **1992**, 148, 478–482.
65. Coelho, P.H.D.S.L.; Marchesin, M.S.; Morales, A.R.; Bartoli, J.R. "Electrical percolation, morphological and dispersion properties of MWCNT/PMMA nanocomposites", *Materials Research*. **2014**, 17, 127–132.
66. Haik, M.Y.; Ayesh, A.I.; Abdulrehman, T.; Haik, Y. "Novel organic memory devices using Au-Pt-Ag nanoparticles as charge storage elements", *Materials Letters*. **2014**, 124(0): 67–72.
67. Liu, X.; Krücker, J.; Zheng, G.; Schubert, D.W. "Mapping the electrical conductivity of poly (methyl methacrylate)/carbon black composites prior to and after shear", *ACS Applied Materials & Interfaces* **2013**, 5(18), 8857–8860.
68. Biswas, B.; Chakraborty, A.K.; Majumder, M.; Chowdhury, A.; Sanyal, M.K.; Mallik, B. "Surface electrical conductivity of poly(methyl methacrylate) thin films: Observation of conductivity switching", *Synthetic Metals* **2012**, 161(23–24), 2632–2637.
69. Hwang, Y.; Je, Y.; Farrar, D.; West, J.E.; Yu, S.M.; Moon, W. "Piezoelectric properties of polypeptide-PMMA molecular composites fabricated by contact charging", *Polymer* **2011**, 52(13), 2723–2728.
70. Hallinan, D.T.; Balsara, N.P. Polymer Electrolytes, in *Annual Review of Materials Research*, Vol 43, D.R. Clarke, Ed.; 2013, pp. 503–525.
71. Flora, X.H.; Ulaganathan, M.; Rajendran, S. "Role of different plasticizers in li-ion conducting poly(acrylonitrile)-poly(methyl methacrylate) hybrid polymer electrolyte", *International Journal of Polymeric Materials and Polymeric Biomaterials* **2013**, 62(14), 737–742.
72. Zhou, L.; Wu, N.; Cao, Q.; Jing, B.; Wang, X.; Wang, Q.; Kuang, H. "A novel electrospun PVDF/PMMA gel polymer electrolyte with in situ TiO₂ for Li-ion batteries", *Solid State Ionics* **2013**, 249–250(0), 93–97.
73. Kufian, M.Z.; Aziz, M.F.; Shukur, M.F.; Rahim, A.S.; Arif, N.E.; Shuhaimi, N.E.A.; Majid, S. R.; Yahya, R.; Arof, A.K. "PMMA – LiBOB gel electrolyte for application in lithium ion batteries", *Solid state Ionic* **2012**, 208, 36–42.
74. Ali, A.M.M.; Subban, R.H.Y.; Bahron, H.; Yahya, M.Z.A.; Kamisan, A.S. "Investigation on modified natural rubber gel polymer electrolytes for lithium polymer battery", *Journal of Power Sources* **2013**, 244, 636–640.
75. Ramesh, S.; Liew, C.-W.; Ramesh, K. "Evaluation and investigation on the effect of ionic liquid onto PMMA-PVC gel polymer blend electrolytes", *Journal of Non-Crystalline Solids* **2011**, 357(10), 2132–2138.
76. Deka, M.; Kumar, A. "Enhanced electrical and electrochemical properties of PMMA –clay nanocomposite gel polymer electrolytes", *Electrochimica Acta* **2010**, 55, 1836–1842.
77. Sharma, J.; Sekhon, S. "Nanodispersed polymer gel electrolytes: Conductivity modification with the addition of PMMA and fumed silica", *Solid State Ionics*, **2007**, 178(5–6), 439–445.
78. Girish, K.G.; Munichandraiah, N. "Poly(methylmethacrylate)—magnesium triflate gel polymer electrolyte for solid state magnesium battery application", *Electrochimica Acta* **2002**, 47 (7), 1013–1022.

79. Agnihotry, S.A.; Pradeep, P.; Sekhon, S.S. "PMMA based gel electrolyte for EC smart windows", *Electrochimica Acta* **1999**, 44(18), 3121–3126.
80. Douglas, A.; Gregory, A.L.; Bruce, K.G. "An electrostatic microvalve for pneumatic control of microfluidic systems". *Journal of Micromechanics and Microengineering* **2012**, 22(2), 1–9.
81. Tan, H.Y.; Loke, W.K.; Nguyen, N.-T. "A reliable method for bonding polydimethylsiloxane (PDMS) to polymethylmethacrylate (PMMA) and its application in micropumps", *Sensors and Actuators B: Chemical* **2010**, 151(1), 133–139.
82. Haibao, L.; Wei Min, H.; Xue Lian, W.; Yu Chun, G.; Fan, Z.; Yong, Z.; Junfeng, G. "Heating/ethanol-response of poly methyl methacrylate (PMMA) with gradient pre-deformation and potential temperature sensor and anti-counterfeit applications", *Smart Materials and Structures* **2014**, 23(6), 1–14.
83. Parambath Kootery, K.; Jiang, H.; Kolusheva, S.; Vinod, T.P.; Ritenberg, M.; Zeiri, L.; Volinsky, R.; Malferrari, D.; Galletti, P.; Tagliavini, E.; Jelinek, R. "Poly(methyl methacrylate)-supported polydiacetylene films: Unique chromatic transitions and molecular sensing", *ACS Applied Materials & Interfaces* **2014**, 6(11), 8613–8620.
84. Mishra, S.K.; Tripathi, S.N.; Choudhary, V.; Gupta, B.D. "SPR based fibre optic ammonia gas sensor utilizing nanocomposite film of PMMA/reduced graphene oxide prepared by in situ polymerization", *Sensors and Actuators B: Chemical* **2014**, 199(0), 190–200.
85. Michael, F.; Kee, S.M.; Sam, K. "A PMMA coated PMN–PT single crystal resonator for sensing chemical agents", *Smart Materials and Structures* **2010**, 19(3), 1–12.
86. Hargsoon, Y.; Jining, X.; Jose, K.A.; Vijay, K.V.; Paul, B. R. "Passive wireless sensors using electrical transition of carbon nanotube junctions in polymer matrix", *Smart Materials and Structures* **2006**, 15(1), S14–S20.
87. Capan, R.; Ray, A.K.; Tanrisever, T.; Hassan, A.K. "Spun thin films of poly(methyl methacrylate) polymer for benzene sensing", *Smart Materials and Structures* **2005**, 14(4), N11–N15.
88. Shen, J.; Li, Z.; Cheng, R.; Luo, Q.; Luo, Y.; Chen, Y.; Chen, X.; Sun, Z.; Huang, S. "Eu3+-doped NaGdF4 nanocrystal down-converting layer for efficient dye-sensitized solar cells", *ACS Applied Materials & Interfaces* **2014**, 6(20), 17454–17462.
89. Yang, H.; Huang, J.; Wu, Z.; Lan, S.; Hao, S.; Lin, J. "The polymer gel electrolyte based on poly(methyl methacrylate) and its application in quasi-solid-state dye-sensitized solar cells", *Materials Chemistry and Physics* **2008**, 110(1), 38–42.
90. Hammam, M.; El-Mansy, M.K.; El-Bashir, S.M.; El-Shaarawy, M.G. "Performance evaluation of thin-film solar concentrators for greenhouse applications", *Desalination* **2007**, 209(1–3), 244–250.
91. Schissel, P.; Jorgensen, G.; Kennedy, C.; Goggin, R. "Silvered-PMMA reflectors", *Solar Energy Materials and Solar Cells* **1994**, 33(2), 183–197.
92. Perween, M.; Parmar, D.B.; Bhadu, G.R.; Srivastava, D.N. "Polymer-graphite composite: a versatile use and throw plastic chip electrode", *Analyst* **2014**, 139(22), 5919–5926.
93. Wang, X.; Wang, P.; Jiang, Y.; Su, Q.; Zheng, J. "Facile surface modification of silica nanoparticles with a combination of noncovalent and covalent methods for composites application", *Composites Science and Technology* **2014**, 104(0), 1–8.
94. Jancar, J.; Douglas, J.F.; Starr, F.W.; Kumar, S.K.; Cassagnau, P.; Lesser, A.J.; Sternstein, S.S.; Buehler, M.J. "Current issues in research on structure–property relationships in polymer nanocomposites", *Polymer* **2010**, 51(15), 3321–3343.
95. Cui, L.; Tarte, N.H.; Woo, S.I. "Synthesis and properties of poly(methyl methacrylate)/carbon nanotube composites covalently integrated through in situ radical polymerization", *Journal of Applied Polymer Science*, **2011**, 119(1), 452–459.
96. Sung, J.H.; Kim, H.S.; Jin, H.-J.; Choi, H.J.; Chin, I.-J. "Nanofibrous membranes prepared by multiwalled carbon nanotube/poly(methyl methacrylate) composites", *Macromolecules* **2004**, 37(26), 9899–9902.